# Registered but Not Anticipated: The Edit-Response Geometry of a Frozen Genomic Foundation Model

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https://github.com/AbdelStark/GenoLeWM

## Abstract

## 1 Introduction

DNA foundation models [9, 15, 20, 21, 25] answer a representational question: *what is the embedding, or the likelihood, of this sequence?* A different question is *interventional*: given a reference genomic window and an explicit edit, what is the *consequence* of that edit in representation space? If a frozen encoder's response to an edit could be *anticipated*—computed from the reference state plus a description of the edit, without re-running the encoder—then a static genomic encoder would become an action-conditioned *world model* [10, 11] whose actions are discrete genomic edits, and whose latent transitions are cheap enough to score variants, roll out haplotypes, and plan edit sets. That is the promise: one expensive encoder pass per reference window, then arbitrarily many cheap latent queries (fig. 1).

This paper measures whether that promise can be kept, on one frozen encoder, and reports that it cannot—for a reason more general than any implementation. Our object of study is the *edit-response geometry* of Carbon-500M [13]: the displacement $\Delta = s_{\mathrm{alt}} - s_{\mathrm{ref}}$ induced in a pooled representation by a single-nucleotide edit. We measure it on $14{,}000$ GRCh38-grounded variants across four pooling widths, and we ask two questions of it in sequence. Does the frozen encoder *register* the edit informatively? And can that registration be *predicted* from the pooled reference state and the action?

The answers dissociate, and the dissociation is the result. The edit is registered richly: $\Delta$ separates ClinVar pathogenic from benign SNVs at AUROC $0.9259$, and this survives every control we could construct—restriction to SNVs, restriction to within a substitution class, and probes that see the reference locus but never the edit. But the registration cannot be anticipated. The best learned predictor of $\Delta$ from $(s_{\mathrm{ref}}, a)$ recovers AUROC $0.6520$, against a ceiling of $0.9259$ from reading the encoder's true response, and against $0.6306$ available from the substitution class alone with no model at all. Adding the pooled reference state to the action buys approximately nothing—and under a linear model it is worse than nothing, scoring $0.5854$ against $0.6079$ for a lookup table that ignores the reference state entirely. **The information exists; it is not where a latent world model needs it to be. There is no shortcut through the pooled latent: you must run the encoder on the edited sequence.**

**The predict-no-change trap.** A second measurement explains why this failure mode is easy to misread, and why we consider the negative result worth reporting rather than quietly abandoning. At global pooling, $\cos(s_{\mathrm{ref}}, s_{\mathrm{alt}}) = 0.99989$: the edited state is cosine-indistinguishable from the reference to four decimal places. Yet AUROC over those same states is $0.9313$. An objective that asks a model to predict $s_{\mathrm{alt}}$ from $s_{\mathrm{ref}}$—which is what the natural JEPA-style formulation asks—is therefore *degenerate*: "predict no change" attains the target to within $10^{-4}$ while carrying none of

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the discriminative signal. Dilution of the edit under pooling is real and strong ($9.4\times$ in relative displacement), but it is *not* information loss. The trap is the metric, not the representation. An experiment that reports a cosine gap under this formulation has measured its own objective, not the encoder.

**This is a measurement paper with a negative headline.** We did not build a working genomic world model, and we do not claim one. We report what a frozen encoder's pooled edit response contains, what it costs to obtain, and what cannot be learned from it. The intended audience is other groups considering the same architecture; the intended saving is their compute. We are also not proposing a variant-effect predictor: section 6 shows the measurement is strong on curated binary labels and *weak* on quantitative function (Spearman $-0.265$ on BRCA2 saturation genome editing), and we report that as a limitation rather than an aside.

**Contributions.**

1. **An edit-response spectroscopy of a frozen genomic FM** (section 4): the displacement induced by a single-base edit, measured across pooling widths on 14,000 GRCh38-grounded variants, with the sensitivity ceiling ($\cos = 0.99989$ at global pooling) and the dissociation between that ceiling and a flat 0.92–0.93 AUROC.
2. **A controlled demonstration that the response is interventional** (section 4.2): the signal is not the mutation spectrum (0.6306 with no geometry), not the exact local sequence context (conditioning on the full trinucleotide context costs 0.0059 AUROC), and not the genomic neighbourhood (0.6392 from a reference-only probe, tracked by a chromosome-held-out probe at 0.6354, so not locus leakage).
3. **A propagation measurement** (section 4.3): the local and distal components of the response point in genuinely different directions ($\cos = 0.4714$), so pooling does not merely rescale a local token swap; the encoder propagates the edit.
4. **The core negative result** (section 5): the pathogenicity-bearing component of the edit response is precisely the component that is *not* predictable from the pooled reference state plus the action, with the failure worsening as the state is pooled more—as an information-bottleneck account predicts.
5. **An explained retraction** (section 7): our own withdrawn `v0.2.1-r1` release had real implementation defects, but we show the deeper reason it could not have worked, and that a correct implementation would have failed too.

**What we do *not* claim.** No clinical, diagnostic, or decision-support claim of any kind. No claim that this is a competitive variant-effect predictor. No claim about token-level world models, which we did not test. No claim of generality across genomic foundation models: this is one encoder, and until it is replicated it is an anecdote about Carbon. We collect these boundaries in section 8.

## 2 Related Work

**Latent world models over frozen features.** The template we test is DINO-WM [27], which learns an action-conditioned latent forward model on top of frozen DINOv2 [22] patch features and plans in latent space by predicting future patch embeddings. Sobal et al. [26] isolate what a latent dynamics model buys over model-free alternatives on reward-free offline data; the Dreamer line [11] and TD-MPC2 [12] establish, respectively, generative and decoder-free latent dynamics for control; Ha and Schmidhuber [10] originate the frozen-encoder plus latent-controller decomposition. Our setting differs in a way that turns out to be decisive. These are *high action-authority* domains: an action moves the state substantially, so "predict-no-change" is a weak baseline and the target signal is rich. A single-nucleotide edit moves a pooled genomic state by $\cos = 0.99989$ at global pooling, so predict-no-change is nearly unbeatable in absolute-state space, and the science has to move into displacement space (section 4). We are not planning; we are measuring the ceiling that any such world model would be built against.

**JEPAs and anti-collapse regularization.** I-JEPA [1] predicts target-block representations from a context block without a pixel decoder, and V-JEPA 2 [2] adds an action-conditioned variant enabling

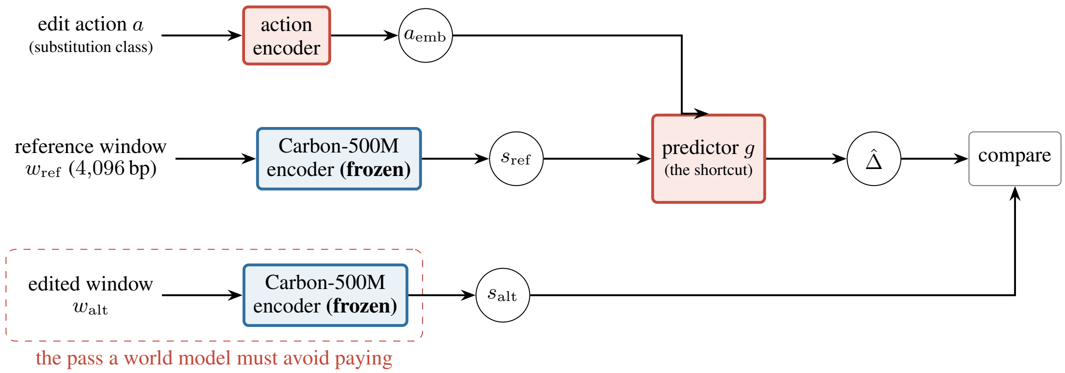


Figure 1: **The object of study.** A latent world model over a frozen genomic encoder freezes Carbon-500M, encodes a reference window once to a pooled state $s_{\text{ref}} \in \mathbb{R}^{1024}$, and trains a small predictor $g$ to anticipate the effect of an edit action $a$ directly in latent space. Its entire value proposition is that the lower (red, dashed) path—running the encoder on the edited window—never has to be taken. We measure the true displacement $\Delta = s_{\text{alt}} - s_{\text{ref}}$ on both paths and ask whether $g(s_{\text{ref}}, a)$ can reproduce it. Section 5 reports that it cannot: the discarded pass is where the pathogenicity signal is created. Section 7 describes the released `v0.2.1-r1` instantiation of this diagram and the implementation defects that made that particular run uninterpretable.

planning from image goals—the closest visual analogue of the predictor we test, with an SNV as the action. LeJEPA [4], by Balestriero and LeCun, proves that among embedding distributions at a fixed covariance budget the isotropic Gaussian uniquely minimizes downstream probe bias, and enforces it with *SIGReg*: random one-dimensional projections with characteristic-function matching, in linear time and memory. **We adopt LeJEPA's isotropic-Gaussian *rationale*, not its estimator.** The regularizer in our own withdrawn release (section 7) was a closed-form Gaussian KL to an isotropic target—a different and weaker mechanism than SIGReg, which we describe here only so that run is characterized accurately. VICReg, Barlow Twins, and DINO supply the surrounding anti-collapse toolbox. The difference from all of these is one of regime rather than method: in vision, the masked target is high-uncertainty and information-rich; here the target is $\approx 99.99\%$ recoverable by copying the input, and the entire signal lives in the residual.

**Genomic foundation models and their variant-effect paradigm.** Carbon is autoregressive and decoder-only with a 6 bp tokenizer, so, like Evo and Evo 2 [7, 21], GPN [5], and HyenaDNA [20], it supports zero-shot $\Delta$ log-likelihood scoring; masked models such as the Nucleotide Transformer [9], DNABERT [15], and Caduceus [25] support pseudo-likelihood analogues. AlphaGenome [3] and supervised track predictors are a contrast class: they score variants by comparing predicted assay tracks, not by a likelihood or a state displacement. Carbon itself has no archival publication; we cite the technical report and pin a model revision (table 7). Every one of these papers reports downstream VEP *accuracy*; none characterizes the intrinsic displacement geometry of a single edit, which is our object.

**Variant-effect prediction: the $\Delta$ log-likelihood and embedding-distance paradigms.** ESM-1v [19] established that a frozen LM's log-likelihood ratio predicts mutation effect with no task supervision, and GPN [5] and Evo [21] carry the identical idea into DNA; AlphaMissense [8] gives broader field context. Our nearest methodological neighbour is **embedding-distance VEP**: BEND [18] and general in-silico-mutagenesis practice already use the distance between reference and alternate embeddings as a variant score. That is precisely the quantity whose geometry we dissect. Prior work *thresholds* $\|\Delta\|$ as a black-box score; we characterize why it behaves as it does and—the part that is new—ask whether it can be *anticipated* rather than measured. The evaluation resources are ClinVar [17], gnomAD [16], TraitGym [6], and MaveDB [23], specifically the BRCA2 saturation genome-editing assay of Sahu et al. [24].

**The gap, and the closest concurrent work.** Across this literature we find no systematic characterization of the pooled-state displacement geometry of a single-nucleotide edit, and no test of whether that displacement is recoverable by an action-conditioned latent predictor. The most direct concurrent neighbour is JEPA-DNA [14], which grounds genomic FMs by predicting masked-segment latent representations and improves linear-probe accuracy. It is a *pretraining objective evaluated by downstream accuracy*; ours is a *measurement of single-edit displacement geometry and its predictability*. The two are complementary, and—if our bottleneck account is right—JEPA-DNA's choice to predict *segment* rather than pooled representations is on the more promising side of exactly the boundary we draw in section 8.

# 3 Method

## 3.1 Frozen encoder and the representation contract

The encoder is Carbon-500M [13], a 500M-parameter decoder-only DNA language model (LLaMA-style, hidden size 1024, 6 bp tokenizer), loaded at pinned revision `5d31d59b` in `bf16` and **frozen throughout**. No encoder parameter is trained anywhere in this paper. Each variant is placed at the centre of a 4,096 bp GRCh38 window. We take layer-20 hidden states $H \in \mathbb{R}^{T\times 1024}$ and pool them to a state $s \in \mathbb{R}^{1024}$.

Three contracts fix the measurement. We state them explicitly because the failure documented in section 7 was caused by leaving them implicit.

**Window.** Reference and edited windows are the same 4,096 bp interval, differing in exactly one base for an SNV. The variant is at the window centre by construction.

**Pooling.** $s^{(r)} = \mathrm{mean}\big(H[c-r : c+r+1]\big)$ for radius $r \in \{8, 64, 256\}$ tokens, where $c$ is the token index of the edited base, derived from the observed DNA-content offset after Carbon's leading control token. Because the tokenizer is 6 bp, these radii are $\pm 48$, $\pm 384$, and $\pm 1536$ bp. We write $r{=}0$ for global-mean pooling over the entire window, which is therefore the *widest* setting, not the narrowest: throughout, the pooling grid is ordered $r{=}8 < r{=}64 < r{=}256 <$ global by width. Reference and edited states are always pooled at the same centre with the same radius.

**Normalization.** All geometry is defined on raw pooled states. We report only quantities that are explicitly scale-aware ($\|\Delta\|_2$) or explicitly scale-free (cos, $\|\Delta\|_2/\|s_{\mathrm{ref}}\|_2$), so that no comparison depends on an unstated normalization convention.

## 3.2 Definitions

For a variant $v$ with reference window $w_{\mathrm{ref}}$ and edited window $w_{\mathrm{alt}} = \mathrm{apply}(v, w_{\mathrm{ref}})$, at pooling width $r$:

$$s_{\mathrm{ref}} = \mathrm{enc}(w_{\mathrm{ref}}), \qquad s_{\mathrm{alt}} = \mathrm{enc}(w_{\mathrm{alt}}), \qquad \Delta \;=\; s_{\mathrm{alt}} - s_{\mathrm{ref}}, \tag{1}$$

and we report the *relative displacement* and the *state cosine*

$$\mathrm{rel}\Delta \;=\; \frac{\|\Delta\|_2}{\|s_{\mathrm{ref}}\|_2}, \qquad \cos(s_{\mathrm{ref}}, s_{\mathrm{alt}}) \;=\; \frac{\langle s_{\mathrm{ref}}, s_{\mathrm{alt}}\rangle}{\|s_{\mathrm{ref}}\|_2\,\|s_{\mathrm{alt}}\|_2}. \tag{2}$$

$\mathrm{rel}\Delta$ measures how far the edit moves the state relative to the state's own scale; $\cos(s_{\mathrm{ref}}, s_{\mathrm{alt}})$ measures how far it moves it in the metric a JEPA-style objective would actually optimize. Section 4 shows these two tell the same story, and that both are misleading about information content.

## 3.3 Cohort

The cohort is 14,000 real GRCh38-grounded variants $\times$ 4 pool radii $=$ 56,000 rows, with 0 skips: ClinVar [17] 1,572 pathogenic / 6,428 benign, and 6,000 BRCA2 saturation genome-editing variants [24] from MaveDB score set `urn:mavedb:00001242-a-1` [23]. Because the initial ClinVar set was not SNV-only, and because indels displace the state far more than substitutions in a way that is aligned with the label, all headline analyses restrict to SNVs: 12,993 SNVs, of which ClinVar contributes 899 pathogenic and 6,094 benign, with 1,007 non-SNV rows excluded and counted. Every one of the 12,993 SNVs was checked against the reference FASTA and matched its recorded reference base (12,993/12,993).

### 3.4 Predictability protocol and its baselines

The world-model question, correctly posed, is not "predict $s_{\text{alt}}$ from $s_{\text{ref}}$"—section 4 shows that target is $\approx 99.99\%$ attained by copying the input. It is: *predict the displacement* $\Delta$ *from the reference state and the action, without encoding the edited window.* That is exactly the efficiency thesis, and it is falsifiable.

The action space is the 12 substitution classes (ordered base-to-base pairs); position is not an action variable because every variant sits at the window centre by construction. We fit predictors $g(s_{\text{ref}}, a) \to \hat{\Delta}$ under 5-fold cross-validation on the 12,993 SNVs, at radius 8—the most favourable, most local pooling width, chosen deliberately so that a failure there is informative about every wider setting. We evaluate two things, and keeping them separate is essential:

**Direction:** $\cos(\hat{\Delta}, \Delta)$, the fidelity of the predicted displacement direction.

**Magnitude:** AUROC of $\|\hat{\Delta}\|_2$ for ClinVar pathogenic vs. benign—whether the predicted response carries the pathogenicity signal that the *true* response carries.

**The baseline ladder.** A negative result is only as good as what it is negative against, so we bracket the learned predictors from below and above. From below: a *global-mean* $\Delta$ predictor that ignores its inputs entirely; a *per-substitution-class-mean lookup*, which is a 12-entry table with no learning; and *the substitution class alone*, scored directly with no displacement at all. From above: the *TRUE* $\Delta$ ceiling—what you get by paying for the encoder pass. A world model is worth building only if it lands meaningfully above the lookup table and near the ceiling.

**Three method notes that constrain interpretation.** First, **a cosine-trained predictor's magnitude is meaningless**: the cosine objective is scale-invariant, so $\|\hat{\Delta}\|$ is unconstrained, and any AUROC computed from it is an artifact of initialization and optimizer path rather than a measurement. We therefore report AUROC($\|\hat{\Delta}\|$) *only* for the MSE-trained predictor, and report the cosine-trained predictor's direction only.

Second, the ridge baseline's $\lambda$ was swept over 1–$10^6$ (best $10^3$) before we concluded anything: an under-regularised failure would prove nothing. Ridge is fit with an intercept. This detail is load-bearing rather than pedantic: an earlier prototype standardised a design matrix containing a constant column, and because that column has zero standard deviation, standardising annihilated it, leaving the model unable to represent $\mathbb{E}[\Delta]$. Every ridge number reported here comes from the corrected fit.

Third, **AUROC is computed with averaged tied ranks**. This matters only for the model-free baselines and it matters a great deal there: the global-mean predictor emits 5 distinct magnitudes and the class-mean lookup 60, so ties dominate their score and a tie-naive implementation misreports them. The MLP-MSE predictor emits 12,993 distinct values with no ties and is unaffected either way—which is why the headline is robust to this choice while the baselines are not. All predictability numbers are those emitted by `tools/research/delta_predictability.py`, using the repository's tie-safe AUROC.

## 4 Edit-Response Spectroscopy

### 4.1 The sensitivity ceiling: dilution is real, information loss is not

Figure 2 sweeps the pooling width. Sensitivity peaks at the narrowest pool and decays monotonically as the pool widens: mean rel$\Delta$ is 0.0759 at $r$=8, 0.0230 at $r$=64, 0.0101 at $r$=256, and 0.0083 at global pooling—a $9.4\times$ swing. The state cosine mirrors it: $\cos(s_{\text{ref}}, s_{\text{alt}})$ is 0.99352 at $r$=8, 0.99925 at $r$=64, 0.99984 at $r$=256, and 0.99989 globally. Under global pooling a single-base edit is invisible to cosine at four decimal places.

The natural inference is that pooling has destroyed the edit. It has not. Over the very same states, ClinVar pathogenic-vs-benign AUROC is *flat*: 0.9259 at $r$=8, 0.9278 at $r$=64, 0.9313 at $r$=256, and 0.9313 globally (table 2). The edit signal survives at every pooling scale. What decays is not the information but its *share of the norm*.

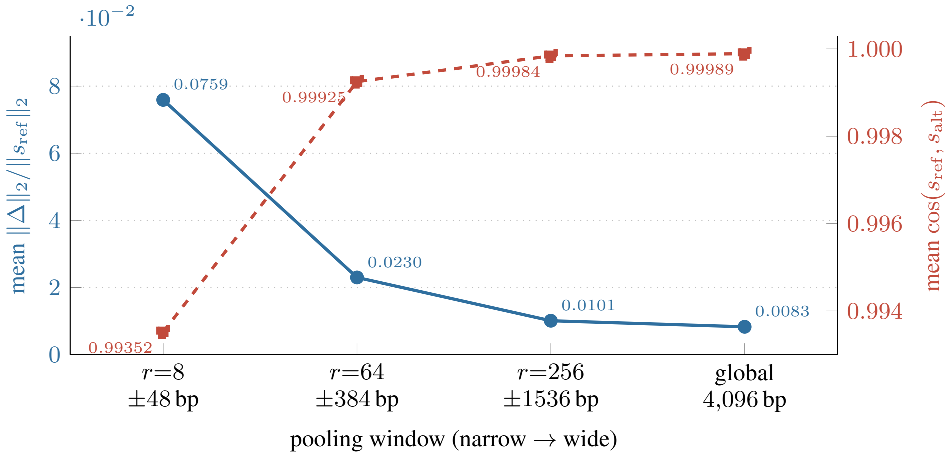


Figure 2: **The sensitivity ceiling: a single-base edit is nearly invisible to cosine at every pooling width, and essentially invisible at global pooling.** Left axis (blue, solid): mean relative displacement $\|\Delta\|_2/\|s_{\mathrm{ref}}\|_2$. Right axis (red, dashed): mean $\cos(s_{\mathrm{ref}}, s_{\mathrm{alt}})$. Both are means over the 14,000-variant cohort. Sensitivity peaks at the narrowest pool ($r$=8, ±48 bp) and falls $9.4\times$ as the pool widens to the full window, where the reference and edited states are cosine-identical to four decimal places (0.99989). *This is not information loss.* Pathogenicity AUROC over the same states is flat at 0.92–0.93 across this entire sweep (fig. 3). A world model trained to predict $s_{\mathrm{alt}}$ from $s_{\mathrm{ref}}$ under a cosine or absolute-state objective is therefore optimising against a target that "predict-no-change" already attains to within $10^{-4}$: the objective is degenerate because the metric is wrong, not because the signal is absent.

This dissociation is the paper's first substantive point, and it retro-diagnoses a whole class of experiments including our own. **"Predict-no-change" wins on cosine because cosine is the wrong metric, not because the information is absent.** A predictor optimizing $1 - \cos(\hat{s}_{\mathrm{alt}}, s_{\mathrm{alt}})$ against a target that the identity function already reaches at 0.99989 is being scored almost entirely on a component of the problem that carries no signal. The objective is degenerate by construction. Displacement space is not a stylistic preference here; it is the only space in which the question is well posed.

### 4.2 The response is interventional, not a confound

An AUROC of 0.93 from a frozen encoder invites suspicion, so we tried to break it (table 2, fig. 3).

**Not indels.** The raw cohort was not SNV-only, and indels displace the state far more than substitutions in a way that correlates with the pathogenic label—an inflation aligned with the outcome. Purging them (1,007 non-SNV rows, counted and excluded) leaves AUROC at 0.9259 at $r$=8. The signal was not the indels.

**Not the mutation spectrum.** Some substitution classes are simply more often pathogenic, and a model could exploit that without any geometry at all. Scoring by substitution class alone—no displacement, no model—gives 0.6306. Restricting the AUROC to comparisons *within* each of the 12 substitution classes (covering $n$=6,993) leaves 0.9202–0.9245 across the sweep. The geometry carries far more than the spectrum, and it carries it after the spectrum is held fixed.

**Not the local sequence context either—the sharpest control.** The 12-class control is weaker than it looks, because it does not condition on local sequence, and a DNA language model's response to a base swap plausibly depends on its immediate neighbours. So we ran the standard mutational-signature decomposition (table 1): condition on the full trinucleotide context—the $5'$ base, the reference base, the $3'$ base—*and* the substitution, then score only within each context. This is the control a referee is most likely to demand, and it is the one that could most plausibly have destroyed the result. It does not. Conditioning on the exact local context costs 0.0059 AUROC: 0.9305 stratified against 0.9364 unstratified, both measured on the identical 5,027 variants

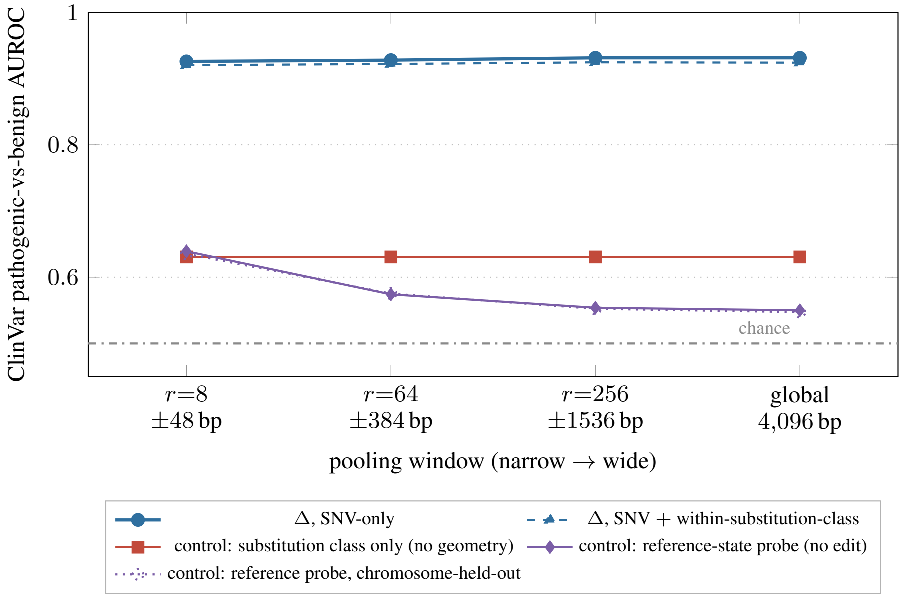


Figure 3: **The edit response is informative and interventional, and it is flat in the pooling width.** The displacement $\Delta$ separates ClinVar pathogenic from benign SNVs at AUROC 0.92–0.93 at every pooling width, including global pooling where $\cos(s_{\mathrm{ref}}, s_{\mathrm{alt}}) = 0.99989$ (fig. 2). Three controls bound the trivial explanations. Scoring variants by *substitution class alone*, with no geometry and no model, reaches 0.6306; restricting the comparison to *within* a substitution class leaves the signal essentially intact (0.9202–0.9245), so the signal is not the mutation spectrum. A linear probe on the reference state alone—which never sees the edit—reaches at most 0.6392, and a chromosome-held-out probe tracks it (0.6354), so the signal is not the genomic neighbourhood and is not locus leakage. What is left is the edit. All numbers in table 2; the sharper trinucleotide-context control is in table 1.

(745 pathogenic) covered by the 37 contexts with at least 10 variants per class. Meanwhile the trinucleotide lookup with zero geometry reaches 0.6618—above the 12-class lookup's 0.6306, as strictly more conditioning information should be, and still far below the geometry. The edit response is not a restatement of mutation chemistry or of local $k$-mer identity.

The apples-to-apples framing is load-bearing and we want to be explicit about it. The stratified 0.9305 exceeds the full-cohort unstratified 0.9259 of table 2, which would look like stratification *improving* the signal. It does not: the 37 retained contexts cover a different and slightly easier subset of the cohort, and against its own subset the stratified score is a small *decrease*. We report the comparison within the shared population because it is the only one that licenses a conclusion.

We flag the limit of this control rather than let it be inferred: trinucleotide context spans ±1 bp, whereas Carbon's tokenizer packs 6 bp into a token. It therefore does not fully control for the exact ref-6-mer → alt-6-mer token pair. The natural next stratification, ±2 bp (5-mers), is too sparse to be powered at 899 pathogenic variants, so we do not report it (section 8).

**Not the genomic neighbourhood.** This was the most serious threat: pathogenic variants cluster in conserved, functionally dense regions, so a probe might recognise the *locus* and never need the edit. A linear probe on $s_{\mathrm{ref}}$ alone—which never observes the edit—reaches 0.6392 at $r$=8, and 0.5500–0.5741 at the wider settings. To exclude locus leakage across cross-validation folds, we re-ran the probe with chromosome-disjoint folds: 0.6354 at $r$=8, tracking the random-fold probe at every radius. There is no same-locus leakage, and the neighbourhood explains at most 0.64 of the 0.93.

What remains, after the spectrum and the neighbourhood are removed and leakage is excluded, is the edit itself. The response is *interventional*.

Table 1: **The trinucleotide-context control: conditioning on the exact local sequence costs essentially nothing.** Radius 8, ClinVar SNVs. This is the standard mutational-signature decomposition: we condition on the $5'$ base, the reference base, the $3'$ base, *and* the substitution, then score only *within* each trinucleotide context. If the edit response were a restatement of mutation chemistry or of local $k$-mer identity, stratifying it away would collapse the AUROC. It costs $0.0059$. The comparison is made *apples-to-apples*, on the identical $5{,}027$ variants that the 37 sufficiently populated contexts cover: comparing the stratified number against the full-cohort $0.9259$ would compare different populations and overstate the result. The two zero-geometry lookups are reported on the full cohort for reference; the trinucleotide lookup beats the 12-class lookup, as strictly more conditioning information should, and both remain far below the geometry. *Caveat:* trinucleotide context spans $\pm 1$ bp while Carbon's tokenizer packs 6 bp per token, so this does not fully control the exact ref-6-mer $\to$ alt-6-mer token pair (section 8).

| Scoring | $n$ | AUROC ↑ |
|---|---|---|
| *Apples-to-apples, on the same subset* ($745$ *pathogenic*) | | |
| Unstratified $\|\Delta\|$ | 5,027 | 0.9364 |
| Within-trinucleotide-context pooled | 5,027 | 0.9305 |
| *cost of conditioning on exact local context* | | $\mathbf{-0.0059}$ |
| *Full-cohort references* | | |
| Unstratified $\|\Delta\|$ | 6,993 | 0.9259 |
| Trinucleotide-context lookup, zero geometry | 6,993 | 0.6618 |
| Substitution-class lookup (12 classes), zero geometry | 6,993 | 0.6306 |

37 trinucleotide contexts retained ($\geq$ 10 variants per class).

### 4.3 The encoder propagates the edit

If a single-base edit merely rewrote its own 6 bp token, then the displacement at a wide pool would be a scaled-down copy of the displacement at a narrow one, and the whole phenomenon would be an uninteresting artifact of averaging. We test this directly by comparing displacement *directions* across pooling widths (fig. 4). Per variant, $\cos(\Delta$ at global pooling, $\Delta$ at $r{=}8)$ has mean $0.4714$, median $0.4721$, and standard deviation $0.1905$; only a fraction $0.004$ of variants exceed $0.9$, and none exceed $0.99$. A pure local token swap would put this distribution at $1.0$. It sits near $0.47$.

Magnitude *ordering*, by contrast, is well preserved: Spearman between $\|\Delta\|$ at global pooling and at $r{=}8$ is $0.8595$. The picture is coherent: *which* variants move the state a lot is stable across scales—which is why AUROC is flat—but *where* they move it is not. The distal response is a genuinely different direction from the local one. This is an attention-mediated, context-dependent response, and the local and distal components independently carry the signal. It also rules out the remaining deflationary reading of section 4.1: the flat AUROC under wide pooling is not a local vector surviving in miniature, it is a distinct global response.

## 5 Predictability: The Core Negative Result

The encoder registers the edit (section 4). A world model does not need the encoder to register the edit; it needs to *anticipate* that registration from the reference state and the action, because anticipating it is the only thing that saves the encoder pass. We now test that, under the protocol of section 3.4, at radius 8—the most favourable setting available.

Table 3 and fig. 5 report the result. Reading the encoder's true response gives AUROC $0.9259$. The best learned predictor of that response, an MLP on $(s_{\mathrm{ref}}, a)$ trained with MSE, gives $0.6520$. The substitution class alone, with no model at all, gives $0.6306$. A 12-entry class-mean lookup table gives $0.6079$. A predictor that ignores its inputs entirely gives $0.4939$, i.e., chance, as it should.

The gap that matters is not the one between $0.6520$ and $0.9259$, though that gap is large. It is the one between $0.6520$ and $0.6306$. **The entire contribution of the learned model—all of it, including the pooled reference state, the nonlinearity, and the training—is the difference between** $0.6520$ **and the** $0.6306$ **you get for free by looking up which base changed into which.**

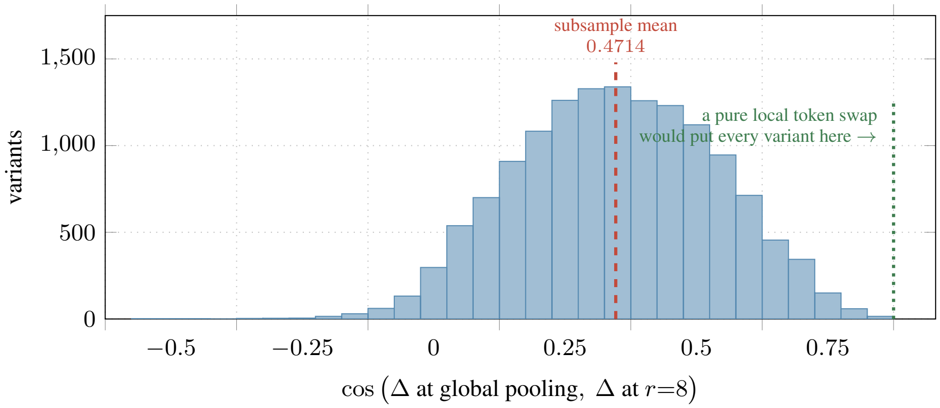


Figure 4: **Carbon propagates the edit; wide pooling is not a rescaling of a local vector.** Per-variant cosine between the displacement measured at global pooling and the displacement measured at $r{=}8$ ($\pm 48$ bp). If a single-base edit merely rewrote its own token, and wider pooling only shrank that same vector, this distribution would be a spike at $1.0$. It is not: on the verified $n{=}2{,}000$ subsample the mean is $0.4714$, the median $0.4721$, and the standard deviation $0.1905$; a fraction of only $0.004$ of variants exceeds $0.9$ and none exceeds $0.99$. The distal response is a genuinely different direction from the local one—an attention-mediated, context-dependent response rather than a token substitution. Magnitude *ordering*, by contrast, is preserved across pooling widths (Spearman $0.8595$ between $\|\Delta\|$ at global pooling and at $r{=}8$), which is why AUROC is flat in fig. 3 while direction is not conserved. Histogram over all $14{,}000$ variants; the quoted statistics are the verified subsample values.

**Conditioning on $s_{\mathrm{ref}}$ is worse than useless.** The stronger statement is available from the linear model, and it is worth stating precisely because it is counterintuitive. Ridge on $(s_{\mathrm{ref}}, a)$ scores AUROC $0.5854$—*below* the $0.6079$ of the class-mean lookup, which ignores $s_{\mathrm{ref}}$ entirely and is a strictly less informed model. Direction agrees: $\cos(\hat{\Delta}, \Delta) = 0.1770$ for ridge against $0.2107$ for the action-only lookup. Giving a linear predictor the pooled reference state does not merely fail to help; it degrades the prediction, because $s_{\mathrm{ref}}$ contributes 1024 dimensions of neighbourhood summary that are near-irrelevant to the response and dilute the one signal (the substitution class) that is relevant. The nonlinear cosine-trained MLP improves direction to $0.3228$, so some structure is recoverable, but the MSE-trained model's magnitude—the component that actually carries the label—sits at $0.1949$ cosine and $0.6520$ AUROC. That the harness is faithful is confirmed by an exact identity: ridge on the action alone reproduces the class-mean lookup to every digit ($\cos = 0.2107$, AUROC $0.6079$), as it must, since one-hot ridge is class means.

**The radius trend confirms the mechanism.** If the cause is an information bottleneck in the pooled state, then predictability should degrade as the state is pooled more, while the ceiling stays flat. It does. At radius $8$ ($\pm 48$ bp, the most local state) the MLP-MSE predictor reaches $0.6520$ against a ceiling of $0.9259$. At global pooling it collapses to $0.5400$—essentially chance—with $\cos(\hat{\Delta}, \Delta) = 0.0575$, against a ceiling that is, if anything, *higher* at $0.9313$. The ceiling is flat in the pooling width; the predictability is not. The wider the pool, the less the response can be anticipated, even though the response stays equally informative. That is precisely the signature of a bottleneck: the pooled state summarises the neighbourhood while discarding the local sequence context—which 6-mer token is being rewritten, and in what immediate surround—that determines the response.

**The conclusion, stated plainly.** The pathogenicity signal in the edit response is exactly the part that is not predictable from the pooled reference state plus the action. A latent world model over pooled frozen genomic FM states cannot recover it—not because the predictor is too small or the optimizer too weak, but because the conditioning information is not in $s_{\mathrm{ref}}$. The efficiency thesis fails at its premise. **There is no shortcut through the pooled latent: you must run the encoder on the edited sequence.**

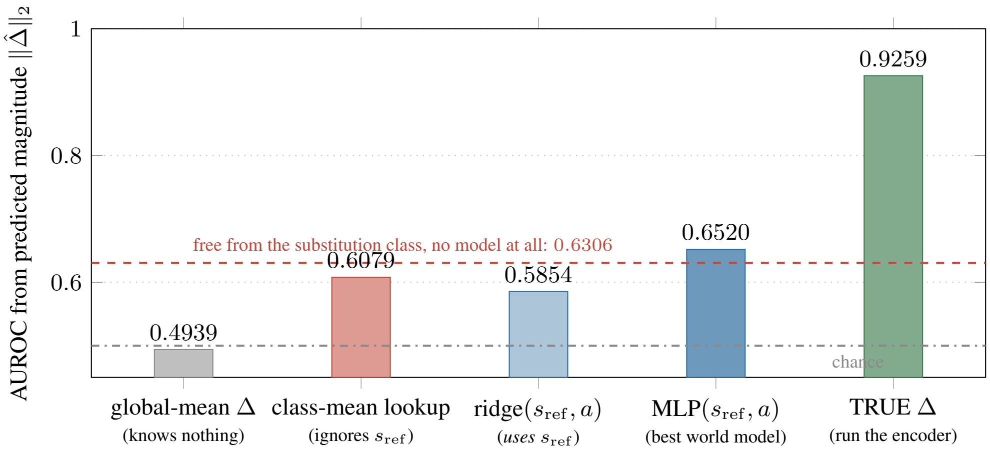


Figure 5: **The core negative result: the pathogenicity-bearing component of the edit response is exactly the component that cannot be anticipated.** All bars are AUROC for ClinVar pathogenic vs. benign, scored by the magnitude of a *predicted* displacement, at radius 8—the most favourable pooling width—under 5-fold cross-validation ($n$=12,993 SNVs). Reading the encoder's true response gives 0.9259 (green). The best learned predictor of that response from the pooled reference state plus the action recovers 0.6520 (blue), barely above the 0.6306 available from the substitution class alone with no model whatsoever (red dashed line). Note the third bar: ridge, which *does* see $s_{\mathrm{ref}}$, scores 0.5854—*below* the 0.6079 class-mean lookup, which does not. Conditioning on the pooled reference state is not merely uninformative here; under a linear model it is actively harmful. The information is registered by the encoder; it is not recoverable from the pooled state it leaves behind. Only the MSE-trained predictor's magnitude is reported here: a cosine-trained predictor's $\|\hat{\Delta}\|$ is scale-free and therefore meaningless as a magnitude (section 3.4).

Table 2: **The edit response is informative, and no trivial confound explains it.** ClinVar pathogenic-vs-benign AUROC from the displacement $\Delta$, swept over pooling width, with every control we could construct from the same artifact. *All rows*: the raw cohort, which is contaminated by indels. *SNV-only*: restricted to the 12,993 single-nucleotide variants (899 pathogenic / 6,094 benign in ClinVar), excluding 1,007 non-SNV rows. *Within substitution class*: additionally scored only within each of the 12 substitution classes (covering $n$=6,993), removing the mutation spectrum. *Substitution only*: the spectrum with no geometry and no model. *Reference probe*: a linear probe on $s_{\mathrm{ref}}$ alone, which never observes the edit. *Chromosome held-out*: the same probe with chromosome-disjoint folds, testing for locus leakage. The pooling grid is ordered by width, not by the radius integer: $r$=0 denotes global-mean pooling over the whole 4,096 bp window.

| | | Edit-response $\Delta$ | | Controls (no edit geometry) | | |
|---|---|---|---|---|---|---|
| Pooling | All rows | SNV-only | Within sub. class | Substitution only | Ref. probe | Chrom. held-out |
| $r$=8 ($\pm$48 bp) | 0.9376 | **0.9259** | 0.9202 | 0.6306 | 0.6392 | 0.6354 |
| $r$=64 ($\pm$384 bp) | 0.9362 | **0.9278** | 0.9220 | 0.6306 | 0.5741 | 0.5753 |
| $r$=256 ($\pm$1536 bp) | 0.9323 | **0.9313** | 0.9245 | 0.6306 | 0.5539 | 0.5523 |
| global (4,096 bp) | 0.9301 | **0.9313** | 0.9240 | 0.6306 | 0.5500 | 0.5473 |

## 6 Baselines and Scope

### 6.1 Quantitative function: where the measurement is weak

The 0.93 AUROC is on curated binary labels, and we do not want it read as a variant-effect result. Table 4 reports the harder test. On the BRCA2 saturation genome-editing assay of Sahu et al. [24] ($n$=6,000, radius 8), Spearman between $\|\Delta\|$ and the measured functional score is $-0.265$ (permutation $p < 0.001$). Transferring the direction learned on ClinVar gives $\rho = -0.155$.

These correlations are statistically real and practically weak, and the contrast with 0.93 is the honest content of this section. Two things differ between the evaluations. ClinVar is ascertainment-biased

Table 3: **The edit response is not anticipatable from the pooled reference state.** Prediction of the displacement $\Delta$ from $(s_{\mathrm{ref}}, a)$ *without* encoding the edited window, at radius 8 (the most favourable pooling width), 5-fold cross-validation, $n$=12,993 SNVs. $\cos(\hat{\Delta}, \Delta)$ measures direction; AUROC($\|\hat{\Delta}\|$) measures whether the predicted magnitude carries the pathogenicity signal that the true magnitude carries. The cosine-trained MLP's magnitude is omitted by construction, not by selection: a cosine objective is scale-invariant, so its $\|\hat{\Delta}\|$ is unconstrained and its AUROC would not be a fair test (section 3.4). Every learned predictor sits between the model-free baselines and the ceiling, far closer to the baselines; ridge, which sees $s_{\mathrm{ref}}$, scores *below* the class-mean lookup, which does not.

| Predictor of $\Delta$ from $(s_{\mathrm{ref}}, a)$ | $\cos(\hat{\Delta}, \Delta) \uparrow$ | AUROC($\|\hat{\Delta}\|$) $\uparrow$ |
|---|---|---|
| *Model-free references* | | |
| Global-mean $\Delta$ (knows nothing) | 0.0493 | 0.4939 |
| Per-substitution-class-mean lookup | 0.2107 | 0.6079 |
| Substitution class alone, no model at all | – | 0.6306 |
| *Learned world models* | | |
| Ridge$(s_{\mathrm{ref}}, a)$, $\lambda$ swept 1–$10^6$ (best $10^3$) | 0.1770 | 0.5854 |
| MLP$(s_{\mathrm{ref}}, a)$, MSE loss | 0.1949 | **0.6520** |
| MLP$(s_{\mathrm{ref}}, a)$, cosine loss | **0.3228** | n/a (scale-free) |
| *Ceiling* | | |
| **TRUE $\Delta$ (run the encoder on the edited window)** | 1.0 | **0.9259** |

Table 4: **Scope of the measurement beyond curated binary labels.** The BRCA2 rows are the honest limit of the result: strong separation of curated pathogenic/benign extremes does *not* transfer to a quantitative functional score within a single gene. This is not a competitive variant-effect-prediction method and we do not present it as one (section 8).

| Evaluation | Quantity | $n$ | Value |
|---|---|---|---|
| ClinVar (curated binary) | AUROC($\|\Delta\|$), $r$=8, SNV-only | 6,993 | 0.9259 |
| BRCA2 sGE (quantitative) | Spearman($\|\Delta\|$, function), $r$=8 | 6,000 | −0.265 |
| BRCA2 sGE (quantitative) | Spearman, ClinVar-learned direction transfer | 6,000 | −0.155 |
| Carbon $\Delta$ log-likelihood | AUROC | – | PENDING |
| Carbon $\Delta$ log-likelihood | Spearman($\|\Delta\|$, $\Delta \log \mathrm{Lik}$) | – | PENDING |

toward clear-cut variants—it is a curated record of what clinicians confidently called, so its positives and negatives are drawn from the tails. BRCA2 sGE, by contrast, spans all SNVs in one gene's exons at fixed context, including the entire ambiguous middle, and asks for a graded quantity rather than a binary one. Separating curated extremes is a much easier problem than ordering a functional gradient. **This is not a competitive variant-effect-prediction method and must not be presented as one.** We report the BRCA2 number because it bounds what the geometry measurement is good for.

### 6.2 Relation to the model's own likelihood

[SECTION PENDING: EXPERIMENT R5.]

## 7 Why the Released Run Failed, and Why a Correct One Would Have Too

This project previously released `v0.2.1-r1`: an action-conditioned latent predictor over a frozen Carbon encoder, trained to predict $\hat{s}_{t+1}$ from $(s_t, a)$, benchmarked on ClinVar, BRCA2, and TraitGym, and published with content-addressed receipts binding every reported number to its bytes. Its results were withdrawn. This section preserves that record, and then makes the point that matters: the defects were real, but they are not the reason the idea could not work.

### 7.1 What was actually broken

A post-release audit established four defects, each verified against the released artifacts (table 6, table 5).

**The normalization contract was ignored.** The configuration set `normalize: true`, but the encoder returned raw pooled states while the predictor always returned a unit-norm vector. On the released rollout slices, mean source/target norms were $33.982/33.595$ (phased) and $29.253/29.089$ (synthetic), against a mean prediction norm of $1.000$. Every $L_2$ loss, surprise score, and rollout distance therefore compared operands differing by a factor of $\approx 30$, and was dominated by scale rather than by content.

**Pooling coordinates were mismatched.** Training encoded source states with no edit locus supplied, which silently triggered global-mean fallback, while targets were pooled around the edit. Source and target states were thus in different coordinate frames. Separately, every "centered" pool omitted Carbon's leading `<dna>` control token and was shifted one hidden token left, sometimes centering the pool on the control token itself.

**The regularizer had no gradient.** The run was labelled `phase2`, but the encoder remained frozen with no LoRA parameters, and its Gaussian KL term was computed only from frozen target states—constant with respect to every trainable parameter, contributing exactly zero gradient. No Phase-2 claim was tested by it.

**The runtime was not self-contained.** The pinned Carbon `tokenizer.py` performed an unpinned, network-capable transitive load of `Qwen/Qwen3-4B-Base`, so hashing the local tokenizer file did not define the tokenizer that actually ran.

Consequently the released $L_2$ losses, surprise scores, $L_2$ rollout metrics, and every conclusion drawn from them do not evaluate the intended method. The released cosine values ($0.289$ and $0.302$ prediction-to-target, versus $0.998$ and $0.991$ source-to-target) remain reproducible historical measurements—cosine is scale-invariant—but they are confounded by mixed-scale, mixed-pooling training and by train/rollout distribution shift, and they establish neither capability nor mechanism. The run used seed 271828, 10,000 optimizer steps, and 80,000 samples; its ClinVar slices were $N{=}16$. We preserve these numbers in table 5 and table 6, and their identity in table 7, for auditability; we withdraw the interpretations.

### 7.2 The lesson the defects do not teach

That audit explains why *that run* was uninterpretable. It is silent on whether the idea was sound. It is tempting—and it was our first reading—to conclude that the engineering was at fault and that a corrected rerun would settle the question. Section 5 says otherwise.

Consider a hypothetical corrected `v0.2.1`: normalization applied consistently, source and target pooled at the same token-layout-derived centre, a gradient-bearing regularizer, a self-contained tokenizer, matched train and rollout distributions, and adequately powered slices. It would have failed anyway, for two compounding reasons that no amount of engineering removes.

**First, its objective was degenerate.** It trained $g(s_t, a) \to \hat{s}_{t+1}$ against a target with $\cos(s_t, s_{t+1}) = 0.99989$ at global pooling (section 4.1). The identity function attains that target to four decimal places. A corrected run would have converged to a near-copy of its input and reported a good-looking loss, and the loss would have measured nothing, because the discriminative content lives in a residual the metric barely weighs. Fixing the normalization bug would have made the numbers *coherent* without making them *informative*.

**Second, and fatally, the conditioning information was absent.** Even after moving to the correctly-posed displacement objective—which is what section 5 does, with a clean implementation on clean data—the best predictor of $\Delta$ from $(s_{\text{ref}}, a)$ recovers $0.6520$ of the $0.9259$ available, barely above the $0.6306$ that requires no model at all. The pooled state is an information bottleneck that discards precisely the local sequence context determining the edit response. This is a property of pooled frozen genomic FM features, not of our optimizer, our architecture, or our bugs. It would have been true of a bug-free implementation. It would have been true of a better-designed one.

Table 5: **Historical variant-effect measurements from the withdrawn `v0.2.1-r1` release; retained for auditability, invalid as a method comparison.** The GenoLeWM score was an $L_2$ residual between unit-norm predictions and raw target states whose norms were about 29–34 (table 6), so these rows do not evaluate the intended normalized-state method. The slices are additionally underpowered ($N$=16 per ClinVar slice, $N$=32 for BRCA2 and TraitGym). $\Delta$ is GenoLeWM − Carbon. Bold marks only the larger historical value.

| Benchmark | Metric | GenoLeWM | Carbon 0-shot | $\Delta$ |
|---|---|---|---|---|
| ClinVar coding | Accuracy ↑ | **0.7500** | 0.6875 | +0.0625 |
| | AUROC ↑ | 0.7344 | **0.9219** | −0.1875 |
| | AP ↑ | 0.8530 | **0.9519** | −0.0989 |
| | Balanced acc. ↑ | **0.7500** | 0.6875 | +0.0625 |
| ClinVar non-coding | Accuracy ↑ | 0.4375 | **0.6875** | −0.2500 |
| | AUROC ↑ | 0.5625 | **0.8750** | −0.3125 |
| | AP ↑ | 0.6055 | **0.9144** | −0.3090 |
| | Balanced acc. ↑ | 0.4375 | **0.6875** | −0.2500 |
| BRCA2 saturation | Spearman $\rho$ ↑ | 0.1492 | **0.4769** | −0.3277 |
| TraitGym Mendelian | Spearman $\rho$ ↑ | −0.0280 | −0.0839 | +0.0559 |

Table 6: **Historical rollout measurements and the state-norm audit that withdrew them (`v0.2.1-r1`).** The lower block is the direct evidence of the normalization defect: the predictor emitted unit-norm vectors while the encoder emitted raw states with norms near 29–34, so the $L_2$ columns compare incommensurable operands. The cosine columns are scale-invariant and remain numerically reproducible, but are confounded by mixed-scale training, mismatched pooling coordinates, and train/rollout distribution shift. Recall@$k$ was 1.0 for both model and baseline at $N$=8 and was non-discriminating. Section 7 argues that repairing all of this would not have rescued the run.

| | Cosine to $s_{t+1}$ ↑ (confounded) | | | $L_2$ to $s_{t+1}$ (invalid) | | |
|---|---|---|---|---|---|---|
| Setting | GenoLeWM | Source-state | $\Delta$ | GenoLeWM | Source-state | $\Delta$ |
| Phased haplotypes | 0.2889 | **0.9978** | −0.7090 | 33.32 | **2.13** | +31.19 |
| Synthetic edit chains | 0.3016 | **0.9912** | −0.6896 | 28.80 | **3.17** | +25.64 |

| Setting | Mean $\|s_t\|_2$ | Mean $\|s_{t+1}\|_2$ | Mean $\|\hat{s}_{t+1}\|_2$ |
|---|---|---|---|
| Phased haplotypes | 33.982 | 33.595 | 1.000 |
| Synthetic edit chains | 29.253 | 29.089 | 1.000 |

This is why we consider the reframe honest rather than convenient. A retraction that says "we had bugs" generalises to nobody. The measurement says something stronger and more useful: this architecture—an action-conditioned latent world model over pooled frozen genomic FM states—cannot deliver its premise, and the reason is information-theoretic rather than engineering. That converts a local embarrassment into a result other groups can act on, and it is the reading the data supports.

## 8 Limitations

We state these as constraints on the claim, not as caveats to it. Several are severe.

**The claim is about *pooled* states only.** We show that the edit response cannot be anticipated from a pooled reference state plus an action. Pooled states are what latent world models actually use, which is why the result bites. But we have *not* shown that a token-level world model is impossible, and nothing here argues that it is. A predictor operating on the full hidden-state sequence retains the local context that pooling discards, and is the natural follow-up. Our result draws a boundary; it does not close the area behind it.

**One encoder.** Every number in this paper comes from Carbon-500M at one revision, one layer, and one window length. Generality across genomic foundation models is untested. **A single-model result is an anecdote about Carbon until it is replicated**, and we ask readers to hold it at exactly that strength. Whether masked models, longer-context models, or single-nucleotide tokenizers show the same bottleneck is an open question our design cannot answer.

**The action space is $12$ substitution classes.** Because every variant sits at the window centre by construction, position carries no information and the action reduces to which base changed into which. A richer action space—position, local sequence context, the identity of the rewritten 6-mer—is not tested here. This cuts against us in a specific and important way: a predictor given the local context we withheld might do better than $0.6520$. We regard this as the most credible route to overturning our negative result, and it is the same route as the token-level follow-up above.

**The $6$-mer token pair is not fully controlled.** Our trinucleotide control (table 1) conditions on $\pm 1$ bp, but Carbon's tokenizer packs 6 bp into a token, so the exact ref-6-mer $\to$ alt-6-mer token-pair confound survives it. The natural fix, a $\pm 2$ bp (5-mer) stratification, is too sparse to be powered at $899$ pathogenic variants, and we do not report an underpowered version of it. The trinucleotide result therefore bounds the $k$-mer explanation without eliminating it.

**ClinVar ascertainment bias.** ClinVar records what was confidently classified, which biases it toward clear-cut variants at the tails of the effect distribution. AUROC $0.9259$ should be read as separation of curated extremes, not as accuracy on the variants that are clinically difficult—which are exactly the ones anybody needs help with.

**BRCA2 is one gene.** The quantitative-function evaluation ($\rho = -0.265$) covers a single gene's exons in a fixed context, from a single assay. It bounds the claim in the direction of weakness, and it should not be extrapolated in either direction to other genes or assays.

**Not a variant-effect method.** We report AUROC against controls to characterize a representation, not to compete on a benchmark. We do not compare against tuned VEP methods, we do not tune ours, and the BRCA2 result indicates it would not compete.

**No clinical claim of any kind.** Nothing in this paper is a clinical, diagnostic, or decision-support tool or claim. No pathogenicity label enters any encoder, and no output here should inform any decision about any person.

## 9 Conclusion

We measured the edit-response geometry of a frozen genomic foundation model and found a clean dissociation. Carbon-500M *registers* a single-base edit richly: the pooled displacement separates ClinVar pathogenic from benign SNVs at AUROC $0.9259$, and it does so interventionally—not via the mutation spectrum ($0.6306$ with no geometry), not via the exact local sequence context (conditioning on the full trinucleotide context costs $0.0059$), not via the genomic neighbourhood ($0.6392$ from a probe that never sees the edit, with no locus leakage), and not via a local token swap (distal and local response directions agree at only $\cos = 0.4714$). The encoder propagates the edit through attention, and the signal survives at every pooling width, even where the edit is cosine-invisible ($0.99989$).

But that registration cannot be *anticipated*. The best learned predictor of the response from the pooled reference state plus the action recovers AUROC $0.6520$ of the $0.9259$ available, against $0.6306$ free from the substitution class alone. The reference state buys almost nothing, and a linear model conditioned on it does worse ($0.5854$) than a lookup table that ignores it ($0.6079$); predictability degrades toward chance as pooling widens while the ceiling stays flat. The pathogenicity-bearing component of the edit response is exactly the unpredictable one.

The consequence for the architecture we set out to build is unambiguous, and negative. An action-conditioned latent world model over pooled frozen genomic FM features cannot deliver its premise, because the premise is that you can skip the encoder pass on the edited sequence, and the encoder pass is where the signal is created. Our own withdrawn `v0.2.1-r1` run had genuine implementation defects, but the defects were not the reason it failed; a correct implementation would have failed too, and now we can say why. The information exists. It is not where a latent world model needs it to be. **There is no shortcut through the pooled latent.**


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## A Notation

| Symbol | Meaning |
|---|---|
| $w_{\mathrm{ref}}, w_{\mathrm{alt}}$ | reference / edited DNA window (4,096 bp, variant at centre) |
| $\mathrm{enc}(\cdot)$ | Carbon-500M (rev. `5d31d59b`), layer 20, mean pool at radius $r$, raw |
| $s_{\mathrm{ref}}, s_{\mathrm{alt}}$ | pooled reference / edited state, $\mathbb{R}^{1024}$ |
| $\Delta = s_{\mathrm{alt}} - s_{\mathrm{ref}}$ | the edit response (displacement); the object of study |
| $\mathrm{rel}\Delta$ | $\lVert\Delta\rVert_2 / \lVert s_{\mathrm{ref}}\rVert_2$, relative displacement |
| $r$ | pool radius in tokens; 6 bp per token, so $r$=8, 64, 256 $\equiv \pm48, \pm384, \pm1536$ bp |
| $r$=0 | global-mean pooling over the whole window (the *widest* setting) |
| $a$ | the action: one of 12 substitution classes |
| $g(s_{\mathrm{ref}}, a) \to \hat{\Delta}$ | the predictor under test; the "shortcut" |
| $n$ | 14,000 variants; 12,993 after SNV restriction; 6,993 ClinVar SNVs |

## B Artifact Identity and Provenance

Table 7: **Artifact identity and provenance.** The upper block identifies the measurement reported in this paper; the lower block identifies the withdrawn release discussed in section 7, retained so that its historical numbers stay bound to the bytes that produced them. The two lineages are disjoint: no result in section 4 or section 5 depends on any `v0.2.1-r1` artifact.

| Field | Value |
|---|---|
| *Edit-response measurement (this paper)* | |
| Encoder | `HuggingFaceBio/Carbon-500M`, rev. `5d31d59b`, `bf16`, frozen |
| Representation | layer 20; mean pool; window 4,096 bp |
| R1 job | `6a565de7b1669a49bf07274b` (A100-large) |
| R1 code commit | `e23fdf9` |
| Analysis commit | `82eb3c2` (`tools/research/edit_response_analysis.py`) |
| Predictability commit | `187a1c5` (`tools/research/delta_predictability.py`) |
| Cohort | 14,000 GRCh38-grounded variants × 4 pool radii = 56,000 rows; 0 skips |
| FASTA ref-base check | 12,993/12,993 SNVs matched |
| Data | `https://huggingface.co/datasets/abdelstark/geno-lewm-edit-response` |
| Data files | `r1-edit-response-e23fdf9/edit_response_embeddings.parquet`,<br>`r1-edit-response-e23fdf9/edit_response_summary.json`,<br>`r1-edit-response-e23fdf9/variants.jsonl` |
| Code | `tools/research/edit_response_spectroscopy.py`,<br>`tools/research/edit_response_analysis.py`,<br>`tools/research/delta_predictability.py`,<br>`tools/jobs/edit_response_run.sh`, `tools/jobs/delta_loglik_run.sh` |
| *Withdrawn release (section 7; historical only)* | |
| Model release | `geno-lewm-v0.2.1-r1` |
| Model id | `sha256:cddb8f3b9671...f5f3ed6af4` |
| Dataset snapshot | `geno-lewm-data-v0.2.1-r1` |
| Commit | `d9b06815cf8e...5aa4154d` |
| Hardware | NVIDIA H200 (Linux x86_64, glibc 2.35) |
| Suite outputs | 28 / 28 verified at the byte level (semantics unverified) |

## C Citation Verification Notes

Bibliography metadata was checked against publisher pages, DOIs, or arXiv where available. The following caveats apply and should be resolved before any camera-ready version.

- **Carbon has no archival publication.** There is a Hugging Face model card and a `tech-report.pdf` in `github.com/huggingface/carbon`; no arXiv or journal version exists. We cite it as a technical report and pin model revision `5d31d59b`. A moving GitHub artifact is a legitimate reviewer objection; the tech report should be archived.
- **LeJEPA attribution and mechanism.** LeJEPA is by Balestriero and LeCun [4], and its estimator is SIGReg (random one-dimensional projections with characteristic-function matching). Our

withdrawn release used a closed-form Gaussian KL inspired by the isotropic-Gaussian rationale; it is *not* SIGReg, and section 2 states this.

- **BRCA2 source disambiguation.** There are two distinct Nature-2025 BRCA2 saturation genome-editing papers. We cite Sahu et al. [24], DOI `10.1038/s41586-024-08349-1`, and *not* the separate study at DOI `10.1038/s41586-024-08388-8`.
- **MaveDB accession.** The score set `urn:mavedb:00001242-a-1` is the identifier recorded by our data pipeline for the BRCA2 sGE scores. Its mapping to Sahu et al. [24] should be confirmed directly on mavedb.org before assertion.
- **Unverified author lists.** JEPA-DNA authors [14] is confirmed as arXiv:2602.17162 (February 2026), but its author list was not captured and the entry currently carries a placeholder. Assran et al. [2], Oquab et al. [22], and Rubin et al. [23] have truncated author lists.
- **Unconfirmed venue.** Zhou et al. [27] is confirmed as arXiv:2411.04983; its ICLR-2025 acceptance is inferred from OpenReview and not confirmed.
- **Unconfirmed DOIs.** The Brixi et al. [7] and Avsec et al. [3] entries cite bioRxiv DOIs, which are solid; their journal DOIs were not confirmed and are not cited.